

# DSC540Week3&4

April 7, 2025

## 1 Activity 3.01

Import required packages for this exercise and read data through Boston Housing file.

Just adding one panda package to ignore warnings

after adding packages calling Boston housing file for this week exercise

```
[12]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import pandas as pd
pd.options.mode.chained_assignment = None # default='warn'
```

```
[13]: df=pd.read_csv("./Boston_housing.csv")
```

Show top 10 records from the Boston housing file

```
[15]: df.head(10)
```

```
[15]:
```

|   | CRIM    | ZN   | INDUS | CHAS | NOX   | RM    | AGE   | DIS    | RAD | TAX | PTRATIO | \ |
|---|---------|------|-------|------|-------|-------|-------|--------|-----|-----|---------|---|
| 0 | 0.00632 | 18.0 | 2.31  | 0    | 0.538 | 6.575 | 65.2  | 4.0900 | 1   | 296 | 15.3    |   |
| 1 | 0.02731 | 0.0  | 7.07  | 0    | 0.469 | 6.421 | 78.9  | 4.9671 | 2   | 242 | 17.8    |   |
| 2 | 0.02729 | 0.0  | 7.07  | 0    | 0.469 | 7.185 | 61.1  | 4.9671 | 2   | 242 | 17.8    |   |
| 3 | 0.03237 | 0.0  | 2.18  | 0    | 0.458 | 6.998 | 45.8  | 6.0622 | 3   | 222 | 18.7    |   |
| 4 | 0.06905 | 0.0  | 2.18  | 0    | 0.458 | 7.147 | 54.2  | 6.0622 | 3   | 222 | 18.7    |   |
| 5 | 0.02985 | 0.0  | 2.18  | 0    | 0.458 | 6.430 | 58.7  | 6.0622 | 3   | 222 | 18.7    |   |
| 6 | 0.08829 | 12.5 | 7.87  | 0    | 0.524 | 6.012 | 66.6  | 5.5605 | 5   | 311 | 15.2    |   |
| 7 | 0.14455 | 12.5 | 7.87  | 0    | 0.524 | 6.172 | 96.1  | 5.9505 | 5   | 311 | 15.2    |   |
| 8 | 0.21124 | 12.5 | 7.87  | 0    | 0.524 | 5.631 | 100.0 | 6.0821 | 5   | 311 | 15.2    |   |
| 9 | 0.17004 | 12.5 | 7.87  | 0    | 0.524 | 6.004 | 85.9  | 6.5921 | 5   | 311 | 15.2    |   |

|   | B      | LSTAT | PRICE |
|---|--------|-------|-------|
| 0 | 396.90 | 4.98  | 24.0  |
| 1 | 396.90 | 9.14  | 21.6  |
| 2 | 392.83 | 4.03  | 34.7  |
| 3 | 394.63 | 2.94  | 33.4  |
| 4 | 396.90 | 5.33  | 36.2  |

```

5  394.12   5.21   28.7
6  395.60  12.43   22.9
7  396.90  19.15   27.1
8  386.63  29.93   16.5
9  386.71  17.10   18.9

```

Get the dimension of dataframe

```
[20]: df.shape
```

```
[20]: (506, 14)
```

Get the selected attributes from dataframe and get last 10 records from it.

```
[23]: selectedf = df[['CRIM', 'ZN', 'INDUS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX', 'PTRATIO', 'PRICE']]
```

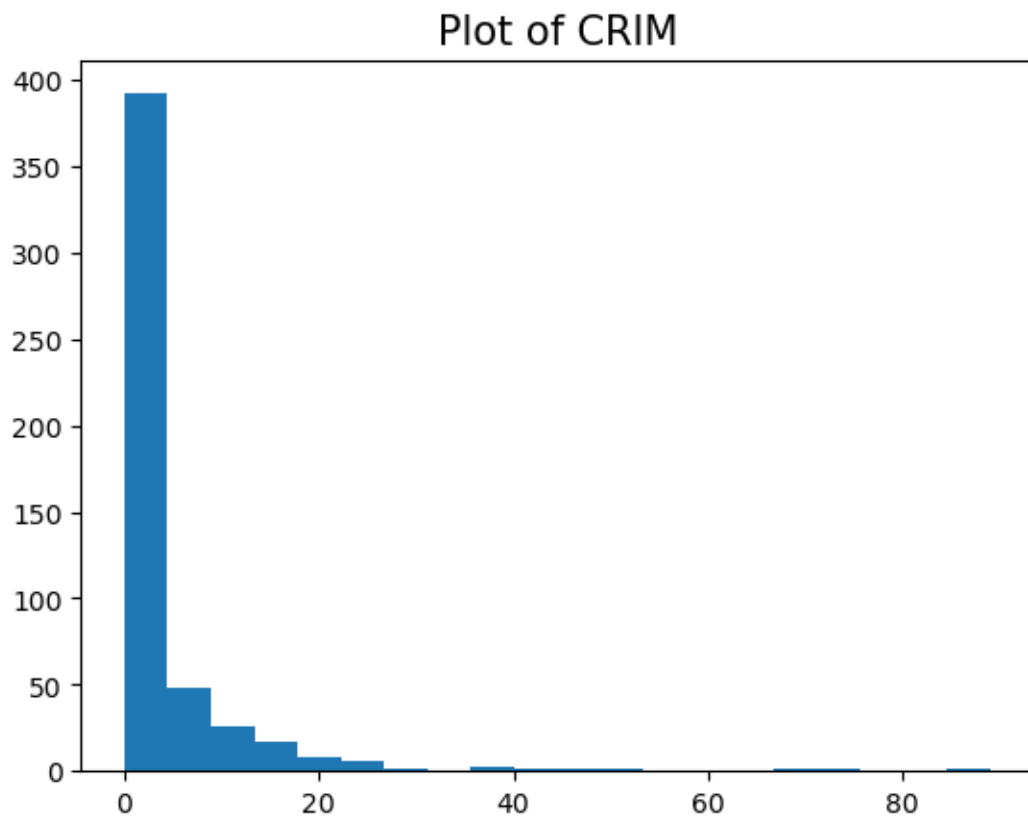
```
[25]: selectedf.tail(10)
```

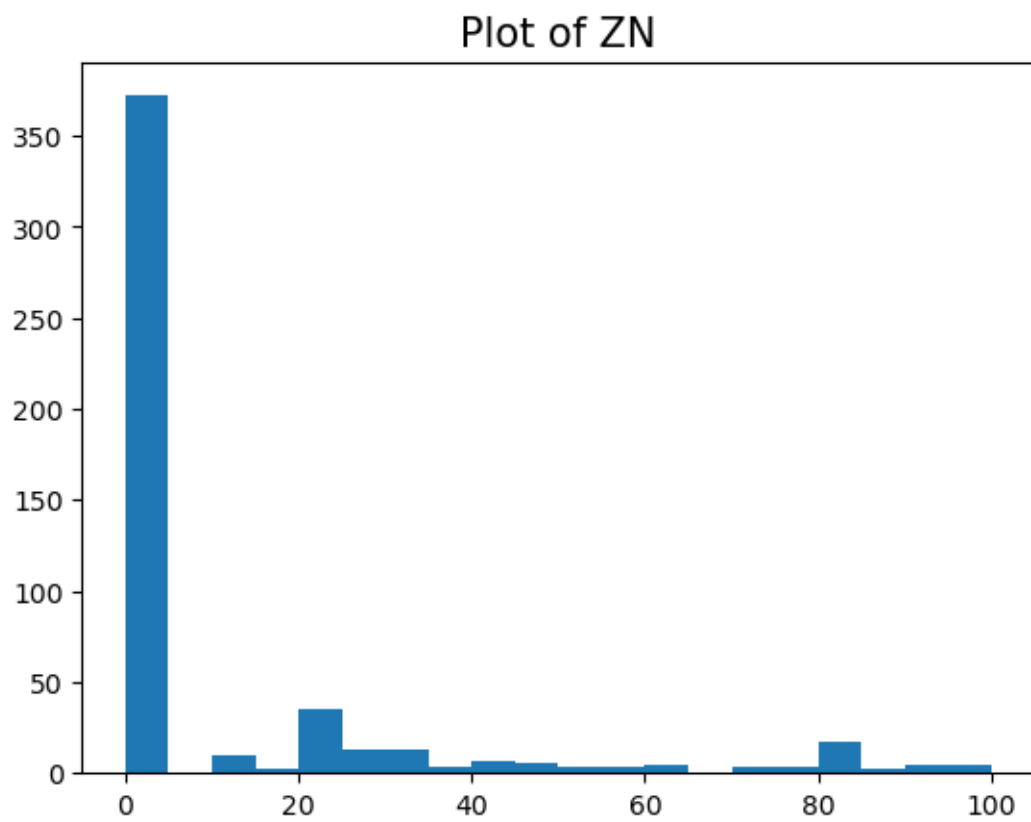
```
[25]:
```

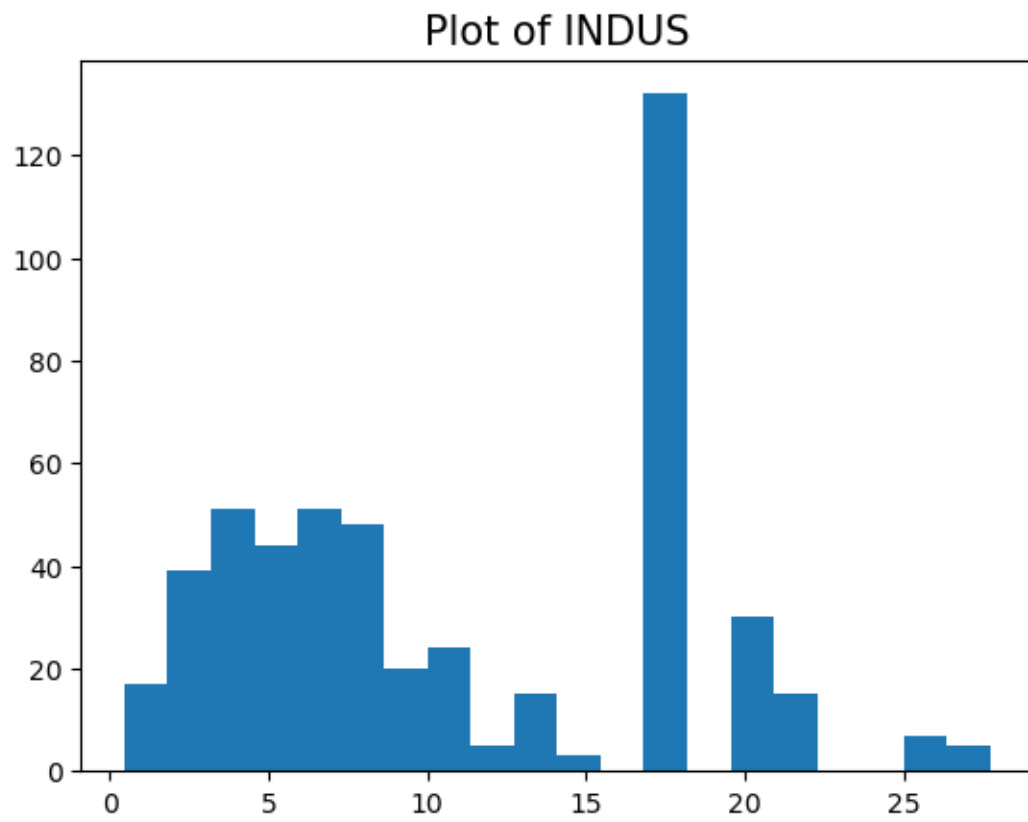
|     | CRIM    | ZN  | INDUS | NOX   | RM    | AGE  | DIS    | RAD | TAX | PTRATIO | PRICE |
|-----|---------|-----|-------|-------|-------|------|--------|-----|-----|---------|-------|
| 496 | 0.28960 | 0.0 | 9.69  | 0.585 | 5.390 | 72.9 | 2.7986 | 6   | 391 | 19.2    | 19.7  |
| 497 | 0.26838 | 0.0 | 9.69  | 0.585 | 5.794 | 70.6 | 2.8927 | 6   | 391 | 19.2    | 18.3  |
| 498 | 0.23912 | 0.0 | 9.69  | 0.585 | 6.019 | 65.3 | 2.4091 | 6   | 391 | 19.2    | 21.2  |
| 499 | 0.17783 | 0.0 | 9.69  | 0.585 | 5.569 | 73.5 | 2.3999 | 6   | 391 | 19.2    | 17.5  |
| 500 | 0.22438 | 0.0 | 9.69  | 0.585 | 6.027 | 79.7 | 2.4982 | 6   | 391 | 19.2    | 16.8  |
| 501 | 0.06263 | 0.0 | 11.93 | 0.573 | 6.593 | 69.1 | 2.4786 | 1   | 273 | 21.0    | 22.4  |
| 502 | 0.04527 | 0.0 | 11.93 | 0.573 | 6.120 | 76.7 | 2.2875 | 1   | 273 | 21.0    | 20.6  |
| 503 | 0.06076 | 0.0 | 11.93 | 0.573 | 6.976 | 91.0 | 2.1675 | 1   | 273 | 21.0    | 23.9  |
| 504 | 0.10959 | 0.0 | 11.93 | 0.573 | 6.794 | 89.3 | 2.3889 | 1   | 273 | 21.0    | 22.0  |
| 505 | 0.04741 | 0.0 | 11.93 | 0.573 | 6.030 | 80.8 | 2.5050 | 1   | 273 | 21.0    | 11.9  |

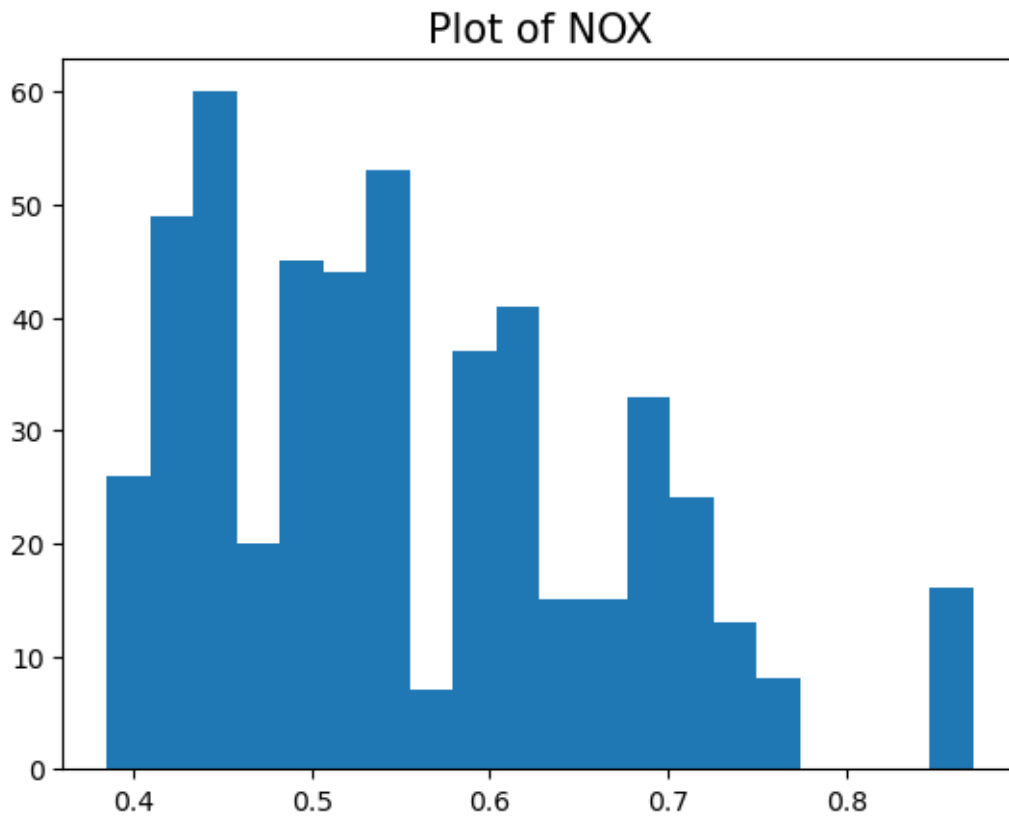
Plot the different histograms based on different attributes from selected dataframe

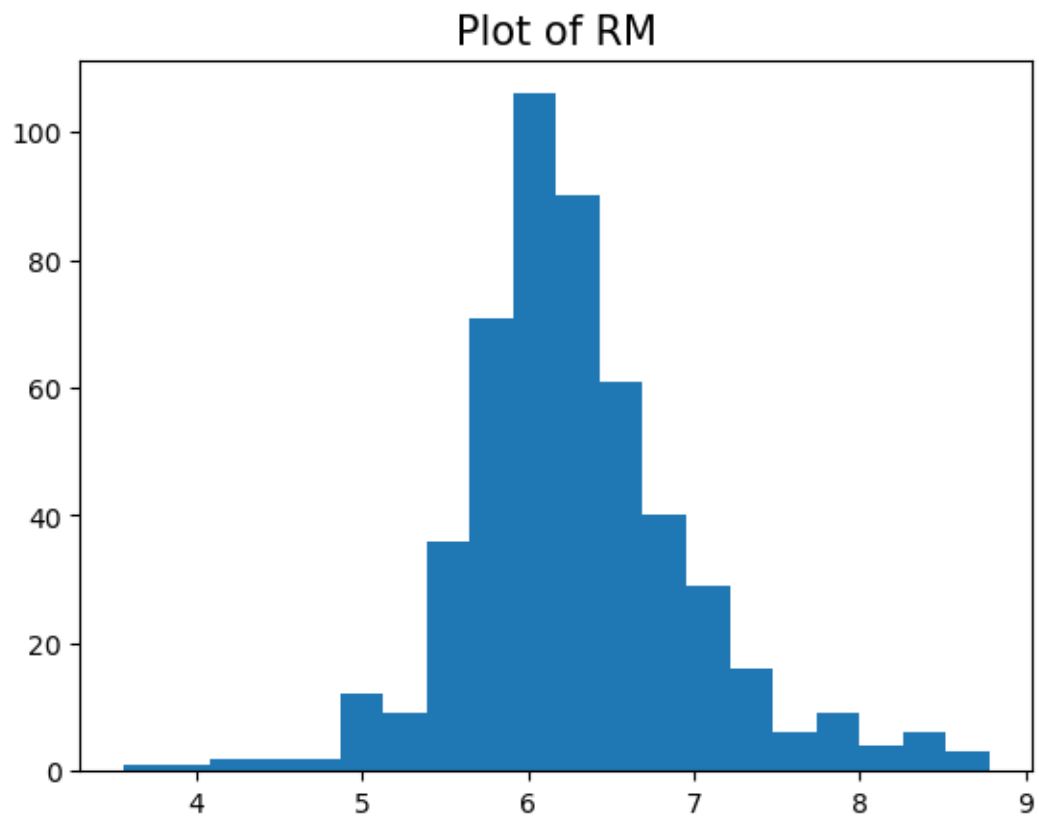
```
[28]: for c in selectedf.columns:
        plt.title("Plot of "+c,fontsize=15)
        plt.hist(selectedf[c],bins=20)
        plt.show()
```

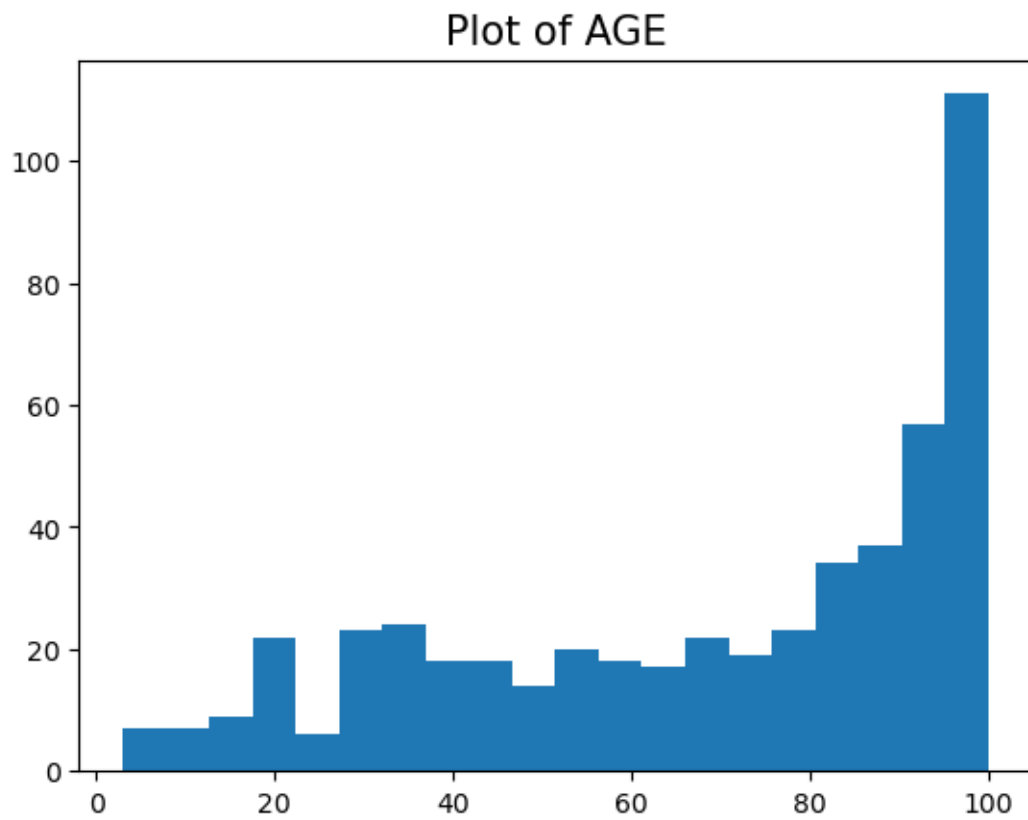






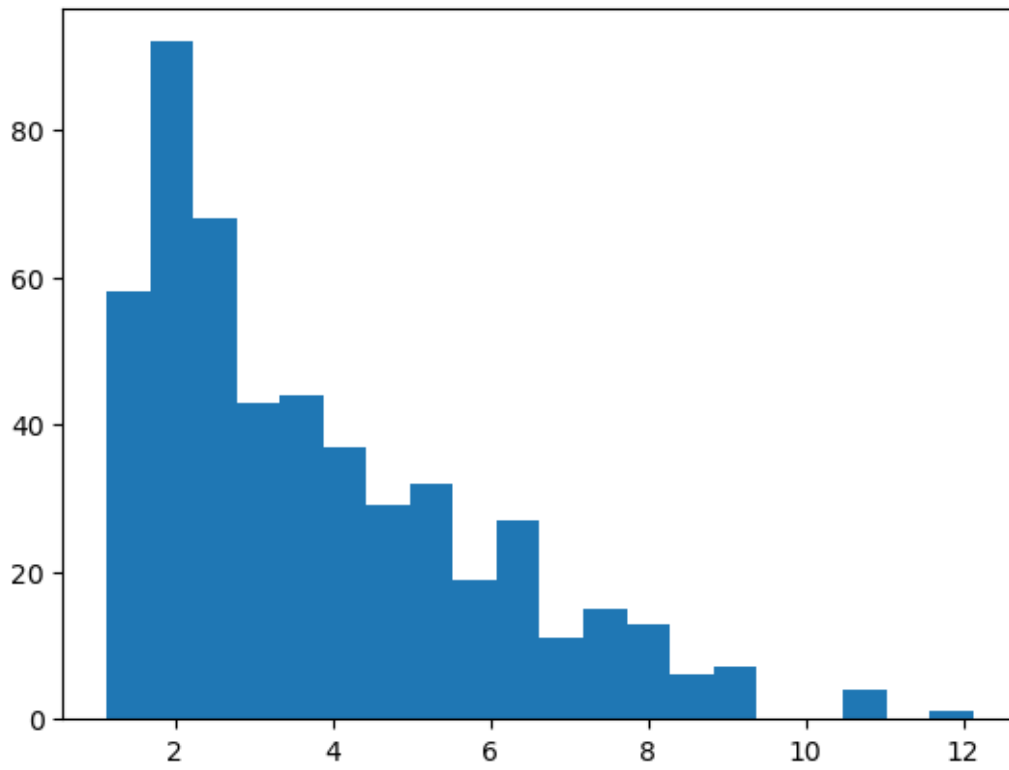


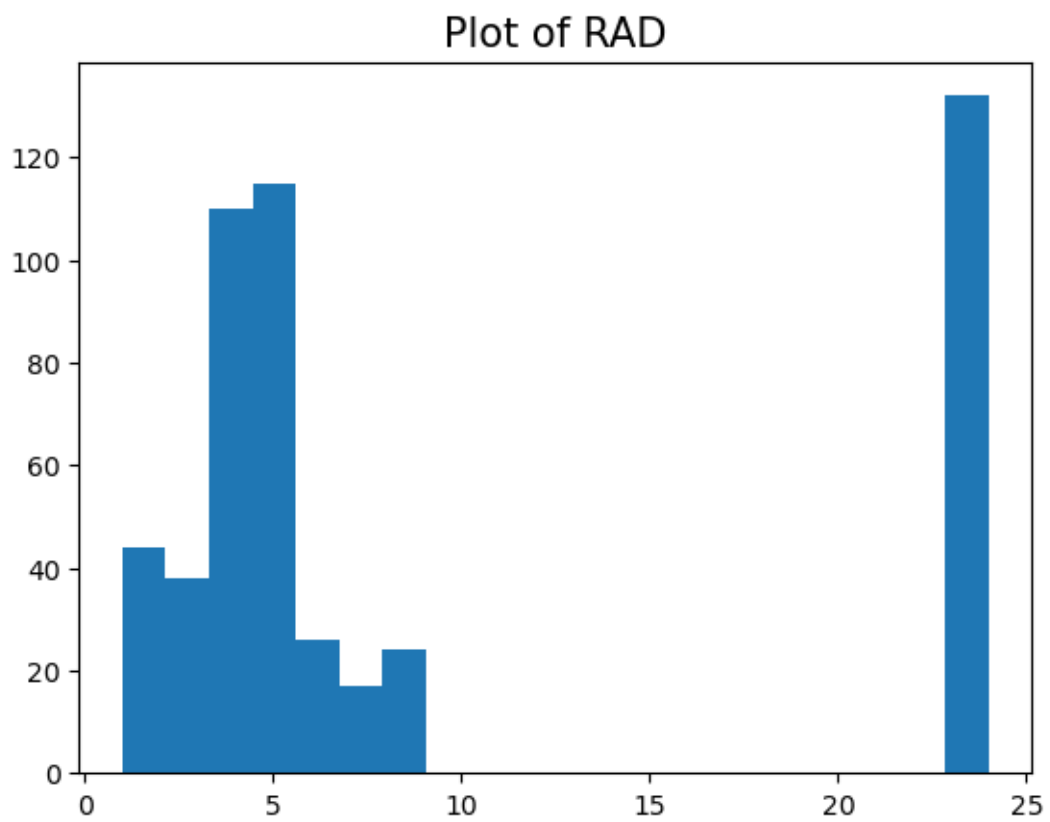


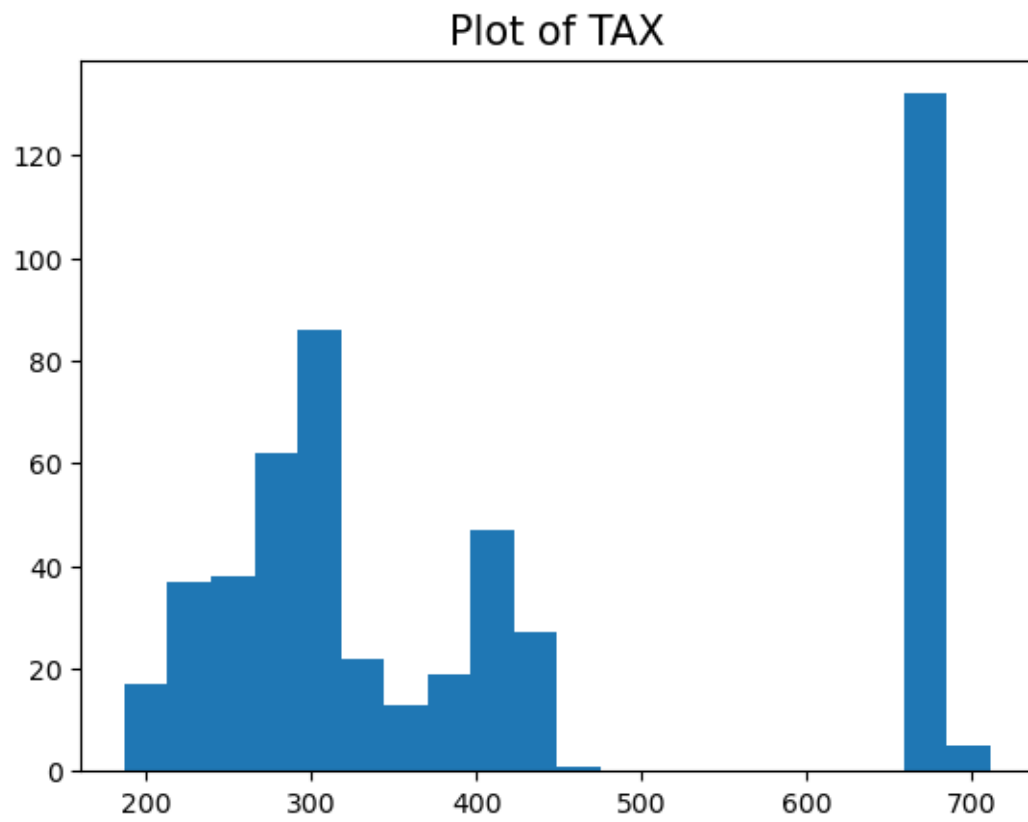


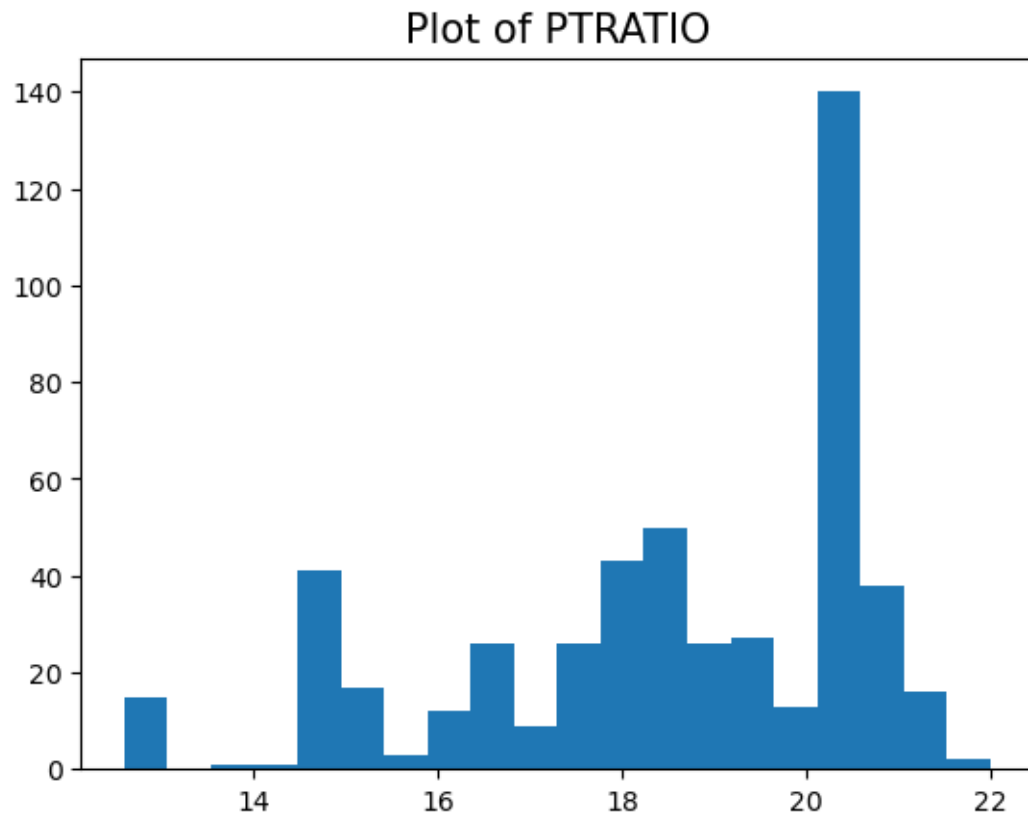


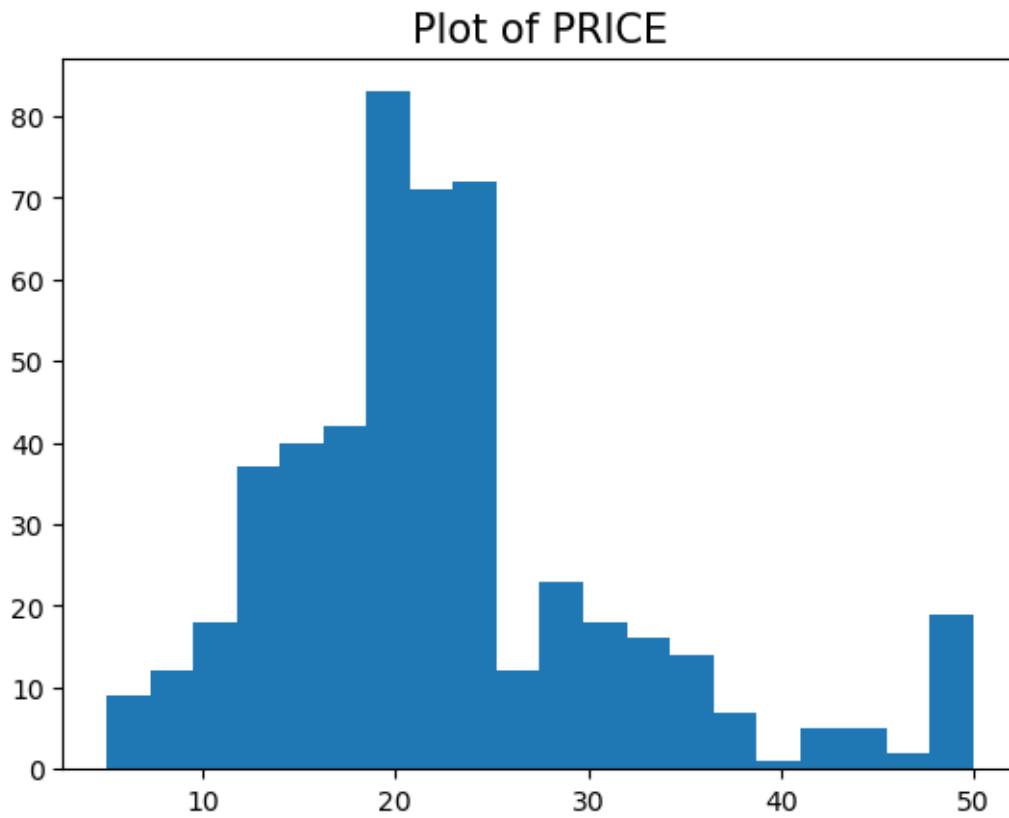
Plot of DIS





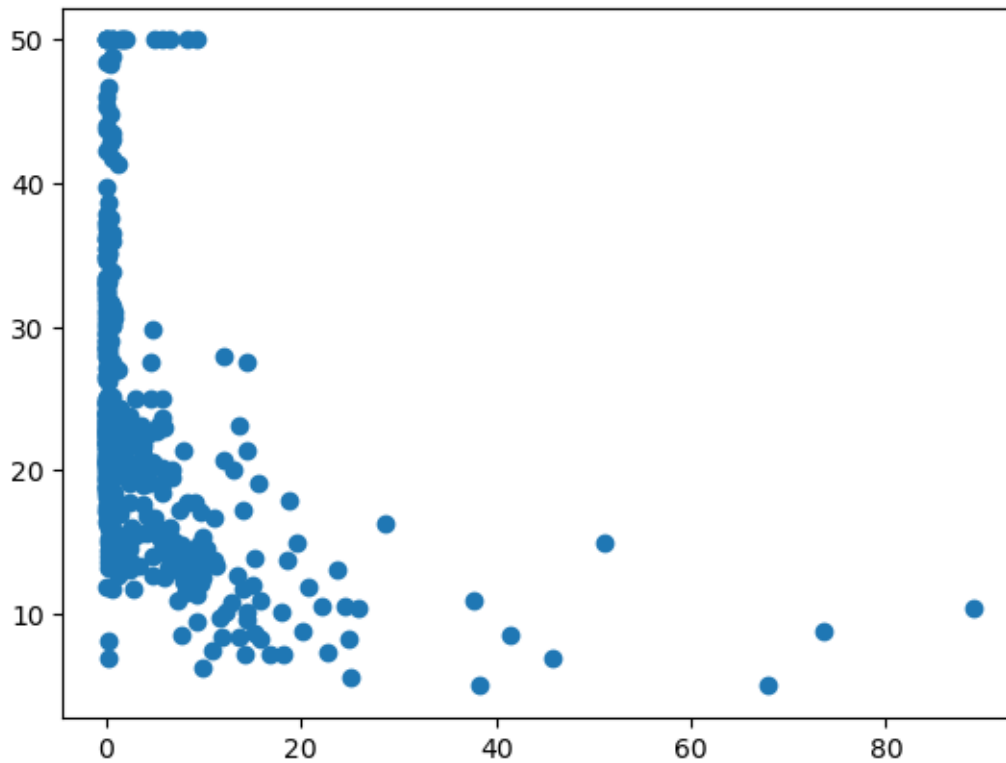






#### 1.0.1 scatter plot for crime rate vs price

```
[30]: plt.scatter(selectedf['CRIM'],selectedf['PRICE'])  
plt.show()
```



Get the type of multiline data

```
[32]: plt.scatter(np.log10(selectedf['CRIM']),selectedf['PRICE'],c='purple')
plt.title("Crime rate (Log) vs. Price plot", fontsize=18)
plt.xlabel("Log of Crime rate",fontsize=15)
plt.ylabel("Price",fontsize=15)
plt.grid(True)
plt.show()
```



Mean values calculation for different attributes

```
[35]: selectedf['RM'].mean()
```

```
[35]: 6.284634387351779
```

```
[39]: selectedf['AGE'].median()
```

```
[39]: 77.5
```

```
[41]: selectedf['DIS'].mean()
```

```
[41]: 3.795042687747036
```

Houses priced below 50,000. So below are the details.

```
[44]: low_price=selectedf['PRICE']<50
      print(low_price)
```

```
0    True
1    True
2    True
```

```

3      True
4      True
...
501    True
502    True
503    True
504    True
505    True
Name: PRICE, Length: 506, dtype: bool

```

Mean values of the low prices <50

```
[47]: low_price.mean()
```

```
[47]: 0.9683794466403162
```

convert that into percentage by multiplying with 100

```
[50]: pcnt=low_price.mean()*100
print("\nPercentage of house with <20,000 price is: ",pcnt)
```

Percentage of house with <20,000 price is: 96.83794466403161

## 2 Activity 4.01

loading adult income file and getting valuable information out of it.

```
[54]: df = pd.read_csv("./adult_income_data.csv")
df.head(10)
```

```
[54]:
```

|      |                  |        |              |    |                       |   |
|------|------------------|--------|--------------|----|-----------------------|---|
| 39   | State-gov        | 77516  | Bachelors    | 13 | Never-married         | \ |
| 0 50 | Self-emp-not-inc | 83311  | Bachelors    | 13 | Married-civ-spouse    |   |
| 1 38 | Private          | 215646 | HS-grad      | 9  | Divorced              |   |
| 2 53 | Private          | 234721 | 11th         | 7  | Married-civ-spouse    |   |
| 3 28 | Private          | 338409 | Bachelors    | 13 | Married-civ-spouse    |   |
| 4 37 | Private          | 284582 | Masters      | 14 | Married-civ-spouse    |   |
| 5 49 | Private          | 160187 | 9th          | 5  | Married-spouse-absent |   |
| 6 52 | Self-emp-not-inc | 209642 | HS-grad      | 9  | Married-civ-spouse    |   |
| 7 31 | Private          | 45781  | Masters      | 14 | Never-married         |   |
| 8 42 | Private          | 159449 | Bachelors    | 13 | Married-civ-spouse    |   |
| 9 37 | Private          | 280464 | Some-college | 10 | Married-civ-spouse    |   |

|   |                   |               |        |      |   |    |               |   |
|---|-------------------|---------------|--------|------|---|----|---------------|---|
|   | Adm-clerical      | Not-in-family | Male   | 2174 | 0 | 40 | United-States | \ |
| 0 | Exec-managerial   | Husband       | Male   | 0    | 0 | 13 | United-States |   |
| 1 | Handlers-cleaners | Not-in-family | Male   | 0    | 0 | 40 | United-States |   |
| 2 | Handlers-cleaners | Husband       | Male   | 0    | 0 | 40 | United-States |   |
| 3 | Prof-specialty    | Wife          | Female | 0    | 0 | 40 | Cuba          |   |
| 4 | Exec-managerial   | Wife          | Female | 0    | 0 | 40 | United-States |   |
| 5 | Other-service     | Not-in-family | Female | 0    | 0 | 16 | Jamaica       |   |



|   |                 |               |        |       |   |    |               |
|---|-----------------|---------------|--------|-------|---|----|---------------|
| 6 | Exec-managerial | Husband       | Male   | 0     | 0 | 45 | United-States |
| 7 | Prof-specialty  | Not-in-family | Female | 14084 | 0 | 50 | United-States |
| 8 | Exec-managerial | Husband       | Male   | 5178  | 0 | 40 | United-States |
| 9 | Exec-managerial | Husband       | Male   | 0     | 0 | 80 | United-States |

|   |       |
|---|-------|
|   | <=50K |
| 0 | <=50K |
| 1 | <=50K |
| 2 | <=50K |
| 3 | <=50K |
| 4 | <=50K |
| 5 | <=50K |
| 6 | >50K  |
| 7 | >50K  |
| 8 | >50K  |
| 9 | >50K  |

```
[56]: names = []
with open('./adult_income_names.txt','r') as f:
    for line in f:
        f.readline()
        var=line.split(":")[0]
        names.append(var)
names
```

```
[56]: ['age',
'workclass',
'fnlwgt',
'education',
'education-num',
'marital-status',
'occupation',
'relationship',
'sex',
'capital-gain',
'capital-loss',
'hours-per-week',
'native-country']
```

```
[58]: names.append('Income')
```

```
[60]: df = pd.read_csv("./adult_income_data.csv",names=names)
df.head()
```

```
[60]:   age      workclass  fnlwgt  education  education-num  \
0   39      State-gov   77516   Bachelors             13
1   50  Self-emp-not-inc   83311   Bachelors             13
```

|   |    |         |        |           |    |
|---|----|---------|--------|-----------|----|
| 2 | 38 | Private | 215646 | HS-grad   | 9  |
| 3 | 53 | Private | 234721 | 11th      | 7  |
| 4 | 28 | Private | 338409 | Bachelors | 13 |

|   | marital-status     | occupation        | relationship  | sex    | \ |
|---|--------------------|-------------------|---------------|--------|---|
| 0 | Never-married      | Adm-clerical      | Not-in-family | Male   |   |
| 1 | Married-civ-spouse | Exec-managerial   | Husband       | Male   |   |
| 2 | Divorced           | Handlers-cleaners | Not-in-family | Male   |   |
| 3 | Married-civ-spouse | Handlers-cleaners | Husband       | Male   |   |
| 4 | Married-civ-spouse | Prof-specialty    | Wife          | Female |   |

|   | capital-gain | capital-loss | hours-per-week | native-country | Income |
|---|--------------|--------------|----------------|----------------|--------|
| 0 | 2174         | 0            | 40             | United-States  | <=50K  |
| 1 | 0            | 0            | 13             | United-States  | <=50K  |
| 2 | 0            | 0            | 40             | United-States  | <=50K  |
| 3 | 0            | 0            | 40             | United-States  | <=50K  |
| 4 | 0            | 0            | 40             | Cuba           | <=50K  |

```
[62]: df.describe()
```

```
[62]:
```

|       | age          | fnlwgt       | education-num | capital-gain | capital-loss | \ |
|-------|--------------|--------------|---------------|--------------|--------------|---|
| count | 32561.000000 | 3.256100e+04 | 32561.000000  | 32561.000000 | 32561.000000 |   |
| mean  | 38.581647    | 1.897784e+05 | 10.080679     | 1077.648844  | 87.303830    |   |
| std   | 13.640433    | 1.055500e+05 | 2.572720      | 7385.292085  | 402.960219   |   |
| min   | 17.000000    | 1.228500e+04 | 1.000000      | 0.000000     | 0.000000     |   |
| 25%   | 28.000000    | 1.178270e+05 | 9.000000      | 0.000000     | 0.000000     |   |
| 50%   | 37.000000    | 1.783560e+05 | 10.000000     | 0.000000     | 0.000000     |   |
| 75%   | 48.000000    | 2.370510e+05 | 12.000000     | 0.000000     | 0.000000     |   |
| max   | 90.000000    | 1.484705e+06 | 16.000000     | 99999.000000 | 4356.000000  |   |

|       | hours-per-week |
|-------|----------------|
| count | 32561.000000   |
| mean  | 40.437456      |
| std   | 12.347429      |
| min   | 1.000000       |
| 25%   | 40.000000      |
| 50%   | 40.000000      |
| 75%   | 45.000000      |
| max   | 99.000000      |

Make a list of all variables with classes

```
[65]: vars_class = ['workclass', 'education', 'marital-status',
                  'occupation', 'relationship', 'sex', 'native-country']
```

```
[67]: for v in vars_class:
        classes=df[v].unique()
        num_classes = df[v].nunique()
```

```

print("There are {} classes in the \"{}\" column. They are: {}".
↪format(num_classes,v,classes))
print("-"*100)

```

There are 9 classes in the "workclass" column. They are: [' State-gov' ' Self-emp-not-inc' ' Private' ' Federal-gov' ' Local-gov' ' ?' ' Self-emp-inc' ' Without-pay' ' Never-worked']

There are 16 classes in the "education" column. They are: [' Bachelors' ' HS-grad' ' 11th' ' Masters' ' 9th' ' Some-college' ' Assoc-acdm' ' Assoc-voc' ' 7th-8th' ' Doctorate' ' Prof-school' ' 5th-6th' ' 10th' ' 1st-4th' ' Preschool' ' 12th']

There are 7 classes in the "marital-status" column. They are: [' Never-married' ' Married-civ-spouse' ' Divorced' ' Married-spouse-absent' ' Separated' ' Married-AF-spouse' ' Widowed']

There are 15 classes in the "occupation" column. They are: [' Adm-clerical' ' Exec-managerial' ' Handlers-cleaners' ' Prof-specialty' ' Other-service' ' Sales' ' Craft-repair' ' Transport-moving' ' Farming-fishing' ' Machine-op-inspct' ' Tech-support' ' ?' ' Protective-serv' ' Armed-Forces' ' Priv-house-serv']

There are 6 classes in the "relationship" column. They are: [' Not-in-family' ' Husband' ' Wife' ' Own-child' ' Unmarried' ' Other-relative']

There are 2 classes in the "sex" column. They are: [' Male' ' Female']

There are 42 classes in the "native-country" column. They are: [' United-States' ' Cuba' ' Jamaica' ' India' ' ?' ' Mexico' ' South' ' Puerto-Rico' ' Honduras' ' England' ' Canada' ' Germany' ' Iran' ' Philippines' ' Italy' ' Poland' ' Columbia' ' Cambodia' ' Thailand' ' Ecuador' ' Laos' ' Taiwan' ' Haiti' ' Portugal' ' Dominican-Republic' ' El-Salvador' ' France' ' Guatemala' ' China' ' Japan' ' Yugoslavia' ' Peru' ' Outlying-US(Guam-USVI-etc)' ' Scotland' ' Trinidad&Tobago' ' Greece' ' Nicaragua' ' Vietnam' ' Hong' ' Ireland' ' Hungary' ' Holand-Netherlands']

```
[69]: df.isnull().sum()
```

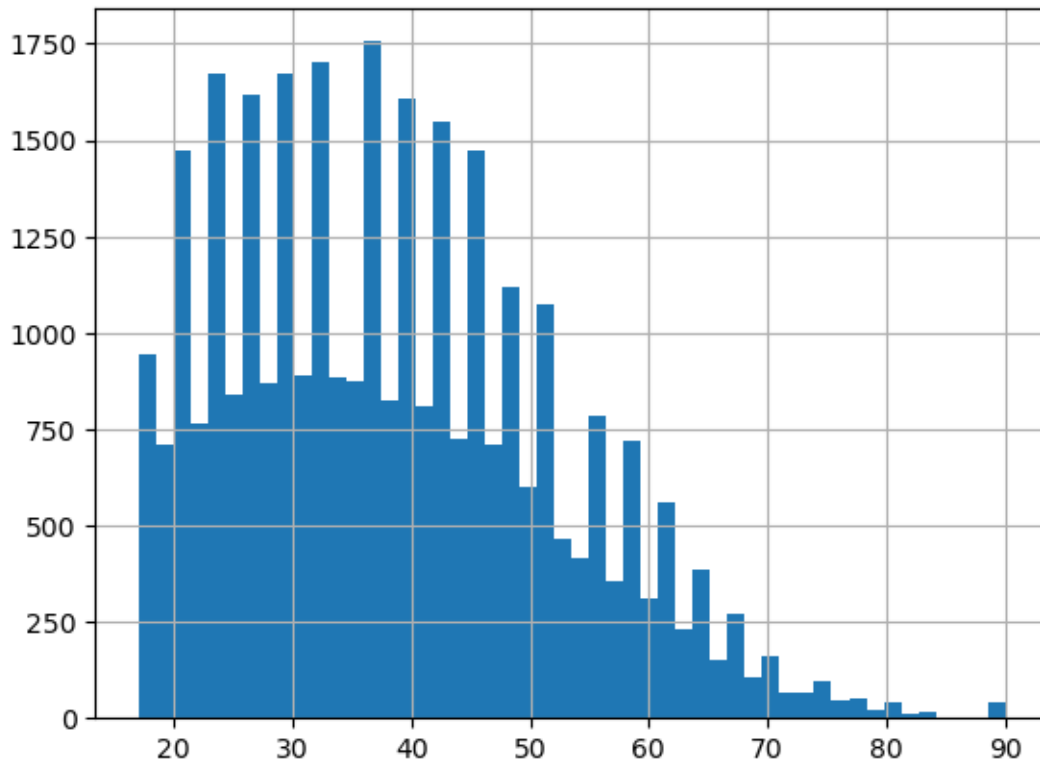
```
[69]: age                0
      workclass         0
      fnlwgt           0
      education         0
      education-num     0
      marital-status    0
      occupation        0
      relationship      0
      sex              0
      capital-gain       0
      capital-loss       0
      hours-per-week     0
      native-country     0
      Income            0
      dtype: int64
```

```
[71]: df_subset = df[['age','education','marital-status']]
      df_subset.head()
```

```
[71]:   age  education  marital-status
0   39  Bachelors  Never-married
1   50  Bachelors  Married-civ-spouse
2   38   HS-grad   Divorced
3   53    11th  Married-civ-spouse
4   28  Bachelors  Married-civ-spouse
```

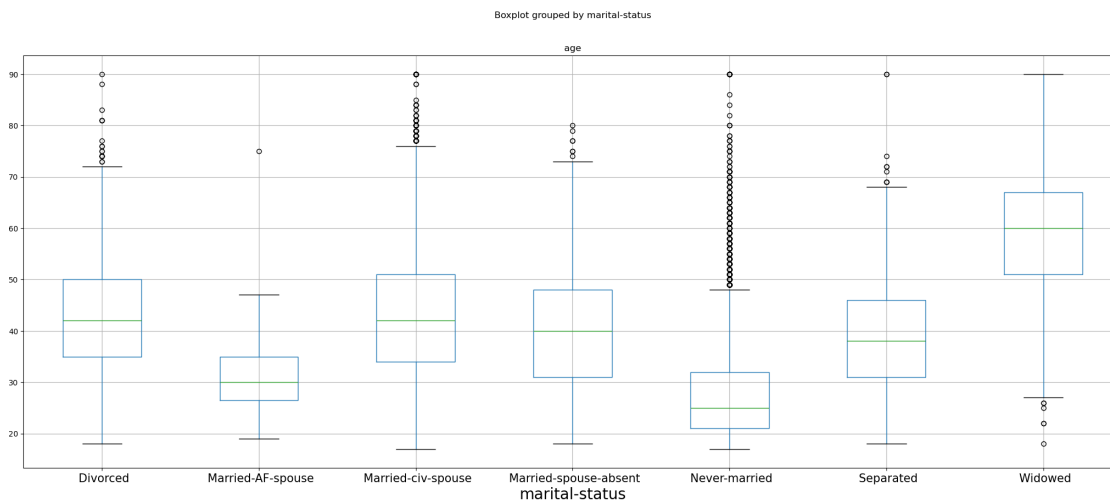
```
[73]: df_subset['age'].hist(bins=50)
```

```
[73]: <Axes: >
```



### 2.0.1 Boxplot showcase of age versus marital-status

```
[76]: df_subset.boxplot(column='age',by='marital-status',figsize=(25,10))
plt.xticks(fontsize=15)
plt.xlabel("marital-status",fontsize=20)
plt.show()
```



strip whitespace function is called for cleaning whitespace from the data.

```
[78]: def strip_whitespace(s):  
      return s.strip()
```

```
[80]: # Education column  
df_subset['education_stripped'] = df['education'].apply(strip_whitespace)  
df_subset['education'] = df_subset['education_stripped']  
df_subset.drop(labels = ['education_stripped'],axis=1,inplace=True)  
  
# Occupation column  
df_subset['occupation_stripped'] = df['occupation'].apply(strip_whitespace)  
df_subset['occupation'] = df_subset['occupation_stripped']  
df_subset.drop(labels = ['occupation_stripped'],axis=1,inplace=True)
```

Conditional clauses and join them by & (AND)

```
[82]: df_filtered=df_subset[(df_subset['age']>=30) & (df_subset['age']<=50)]
```

```
[84]: df_filtered.head(10)
```

```
[84]:
```

|    | age | education    | marital-status        | occupation        |
|----|-----|--------------|-----------------------|-------------------|
| 0  | 39  | Bachelors    | Never-married         | Adm-clerical      |
| 1  | 50  | Bachelors    | Married-civ-spouse    | Exec-managerial   |
| 2  | 38  | HS-grad      | Divorced              | Handlers-cleaners |
| 5  | 37  | Masters      | Married-civ-spouse    | Exec-managerial   |
| 6  | 49  | 9th          | Married-spouse-absent | Other-service     |
| 8  | 31  | Masters      | Never-married         | Prof-specialty    |
| 9  | 42  | Bachelors    | Married-civ-spouse    | Exec-managerial   |
| 10 | 37  | Some-college | Married-civ-spouse    | Exec-managerial   |
| 11 | 30  | Bachelors    | Married-civ-spouse    | Prof-specialty    |
| 13 | 32  | Assoc-acdm   | Never-married         | Sales             |

```
[86]: answer_1=df_filtered.shape[0]  
      answer_1
```

```
[86]: 16390
```

```
[88]: print("There are {} people of age between 30 and 50 in this dataset."  
      ↪format(answer_1))
```

There are 16390 people of age between 30 and 50 in this dataset.

```
[90]: df_subset.groupby('occupation').describe()['age']
```

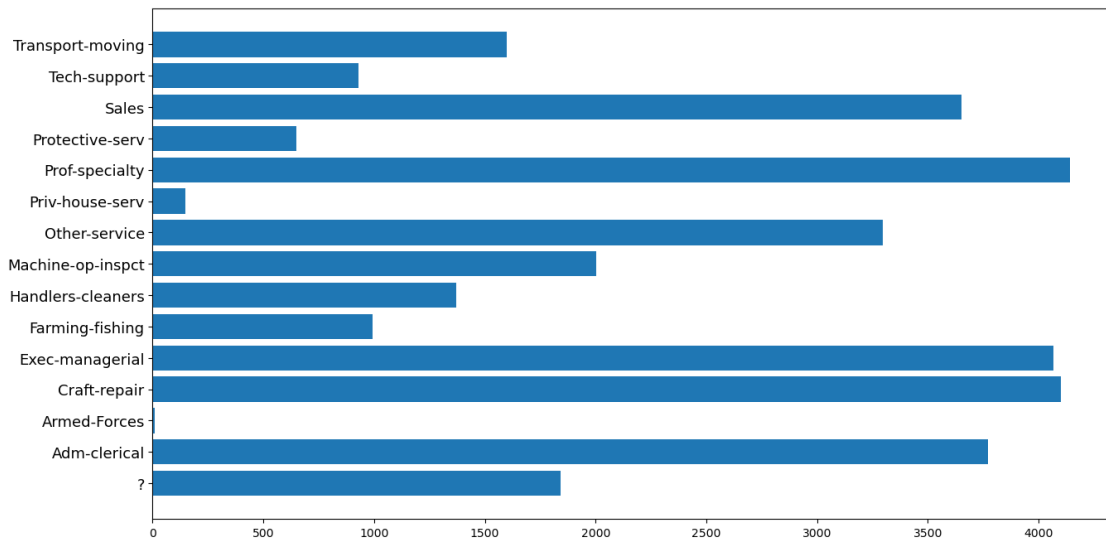
```
[90]:
```

|            | count  | mean      | std       | min  | 25%  | 50%  | 75%  | max  |
|------------|--------|-----------|-----------|------|------|------|------|------|
| occupation |        |           |           |      |      |      |      |      |
| ?          | 1843.0 | 40.882800 | 20.336350 | 17.0 | 21.0 | 35.0 | 61.0 | 90.0 |

|                   |        |           |           |      |      |      |      |      |
|-------------------|--------|-----------|-----------|------|------|------|------|------|
| Adm-clerical      | 3770.0 | 36.964456 | 13.362998 | 17.0 | 26.0 | 35.0 | 46.0 | 90.0 |
| Armed-Forces      | 9.0    | 30.222222 | 8.089774  | 23.0 | 24.0 | 29.0 | 34.0 | 46.0 |
| Craft-repair      | 4099.0 | 39.031471 | 11.606436 | 17.0 | 30.0 | 38.0 | 47.0 | 90.0 |
| Exec-managerial   | 4066.0 | 42.169208 | 11.974548 | 17.0 | 33.0 | 41.0 | 50.0 | 90.0 |
| Farming-fishing   | 994.0  | 41.211268 | 15.070283 | 17.0 | 29.0 | 39.0 | 52.0 | 90.0 |
| Handlers-cleaners | 1370.0 | 32.165693 | 12.372635 | 17.0 | 23.0 | 29.0 | 39.0 | 90.0 |
| Machine-op-inspct | 2002.0 | 37.715285 | 12.068266 | 17.0 | 28.0 | 36.0 | 46.0 | 90.0 |
| Other-service     | 3295.0 | 34.949621 | 14.521508 | 17.0 | 22.0 | 32.0 | 45.0 | 90.0 |
| Priv-house-serv   | 149.0  | 41.724832 | 18.633688 | 17.0 | 24.0 | 40.0 | 57.0 | 81.0 |
| Prof-specialty    | 4140.0 | 40.517633 | 12.016676 | 17.0 | 31.0 | 40.0 | 48.0 | 90.0 |
| Protective-serv   | 649.0  | 38.953775 | 12.822062 | 17.0 | 29.0 | 36.0 | 47.0 | 90.0 |
| Sales             | 3650.0 | 37.353973 | 14.186352 | 17.0 | 25.0 | 35.0 | 47.0 | 90.0 |
| Tech-support      | 928.0  | 37.022629 | 11.316594 | 17.0 | 28.0 | 36.0 | 44.0 | 73.0 |
| Transport-moving  | 1597.0 | 40.197871 | 12.450792 | 17.0 | 30.0 | 39.0 | 49.0 | 90.0 |

```
[92]: occupation_stats= df_subset.groupby('occupation').describe()['age']
```

```
[94]: plt.figure(figsize=(15,8))
plt.barh(y=occupation_stats.index,width=occupation_stats['count'])
plt.yticks(fontsize=13)
plt.show()
```



Following taking sample 10 records for 2 sub dataframes on different set of columns and showing it.

```
[111]: df_1 = df[['age', 'workclass', 'occupation']].sample(10, random_state=101)

df_1.head(10)
```

```
[111]:
```

|       | age | workclass | occupation        |
|-------|-----|-----------|-------------------|
| 22357 | 51  | Private   | Machine-op-inspct |
| 26009 | 19  | Private   | Sales             |
| 20734 | 40  | Private   | Exec-managerial   |
| 17695 | 17  | Private   | Handlers-cleaners |
| 27908 | 61  | Private   | Craft-repair      |
| 27225 | 58  | ?         | ?                 |
| 13108 | 37  | Local-gov | Other-service     |
| 27552 | 22  | Private   | Adm-clerical      |
| 14043 | 22  | Private   | Sales             |
| 30313 | 26  | ?         | ?                 |

```
[98]: df_2 = df[['education', 'occupation']].sample(10, random_state=101)

df_2.head(10)
```

```
[98]:
```

|       | education | occupation        |
|-------|-----------|-------------------|
| 22357 | HS-grad   | Machine-op-inspct |
| 26009 | 11th      | Sales             |
| 20734 | HS-grad   | Exec-managerial   |
| 17695 | 10th      | Handlers-cleaners |
| 27908 | 7th-8th   | Craft-repair      |

Following code merging/joining 2 data frames df1 and df2 on ‘Occupation’ and dropping duplicates records afterwards.

```
[101]: df_merged = pd.merge(df_1, df_2, on='occupation', how='inner').drop_duplicates()

df_merged
```

```
[101]:
```

|   | age | workclass | occupation        | education |
|---|-----|-----------|-------------------|-----------|
| 0 | 51  | Private   | Machine-op-inspct | HS-grad   |
| 1 | 19  | Private   | Sales             | 11th      |
| 2 | 40  | Private   | Exec-managerial   | HS-grad   |
| 3 | 17  | Private   | Handlers-cleaners | 10th      |
| 4 | 61  | Private   | Craft-repair      | 7th-8th   |

Create 2 Pandas Series as per assignment

```
[104]: data_1 = [7.3, -2.5, 3.4, 1.5]
series_1 = pd.Series(data_1)

print("Series 1:")
print(series_1)
```

```
Series 1:
0    7.3
1   -2.5
2    3.4
```



```
3      1.5
dtype: float64
```

```
[106]: data_2 = [-2.1, 3.6, -1.5, 4, 3.1]
       series_2 = pd.Series(data_2)

       print("Series 2:")
       print(series_2)
```

```
Series 2:
0    -2.1
1     3.6
2    -1.5
3     4.0
4     3.1
dtype: float64
```

#### Addition of 2 series [array]

```
[109]: Addseries = series_1 + series_2

       print(Addseries)
```

```
0     5.2
1     1.1
2     1.9
3     5.5
4     NaN
dtype: float64
```

#### Subtraction of 2 series [array]

```
[215]: Subseries = series_1 - series_2

       print(Subseries)
```

```
0     9.4
1    -6.1
2     4.9
3    -2.5
4     NaN
dtype: float64
```

```
[ ]:
```